

WEEKLY LAB MEETING
IBENS ZEBRAFISH NEURO-ETHOLOGY
LNC GROUP FOR NEURAL THEORY

TUESDAY, JANUARY 12TH 2016

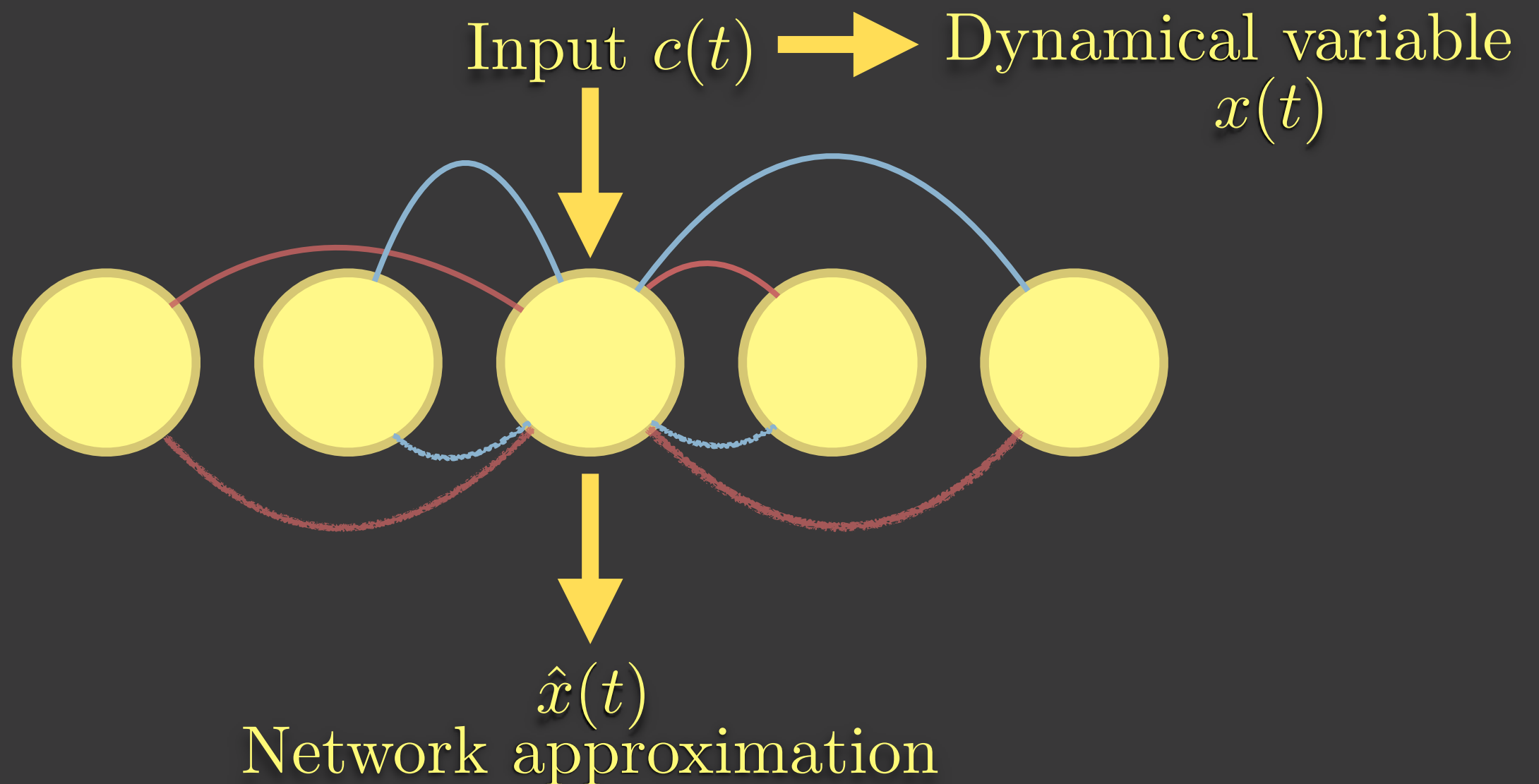
PREDICTIVE CODING IN BALANCED SPIKING NETWORKS



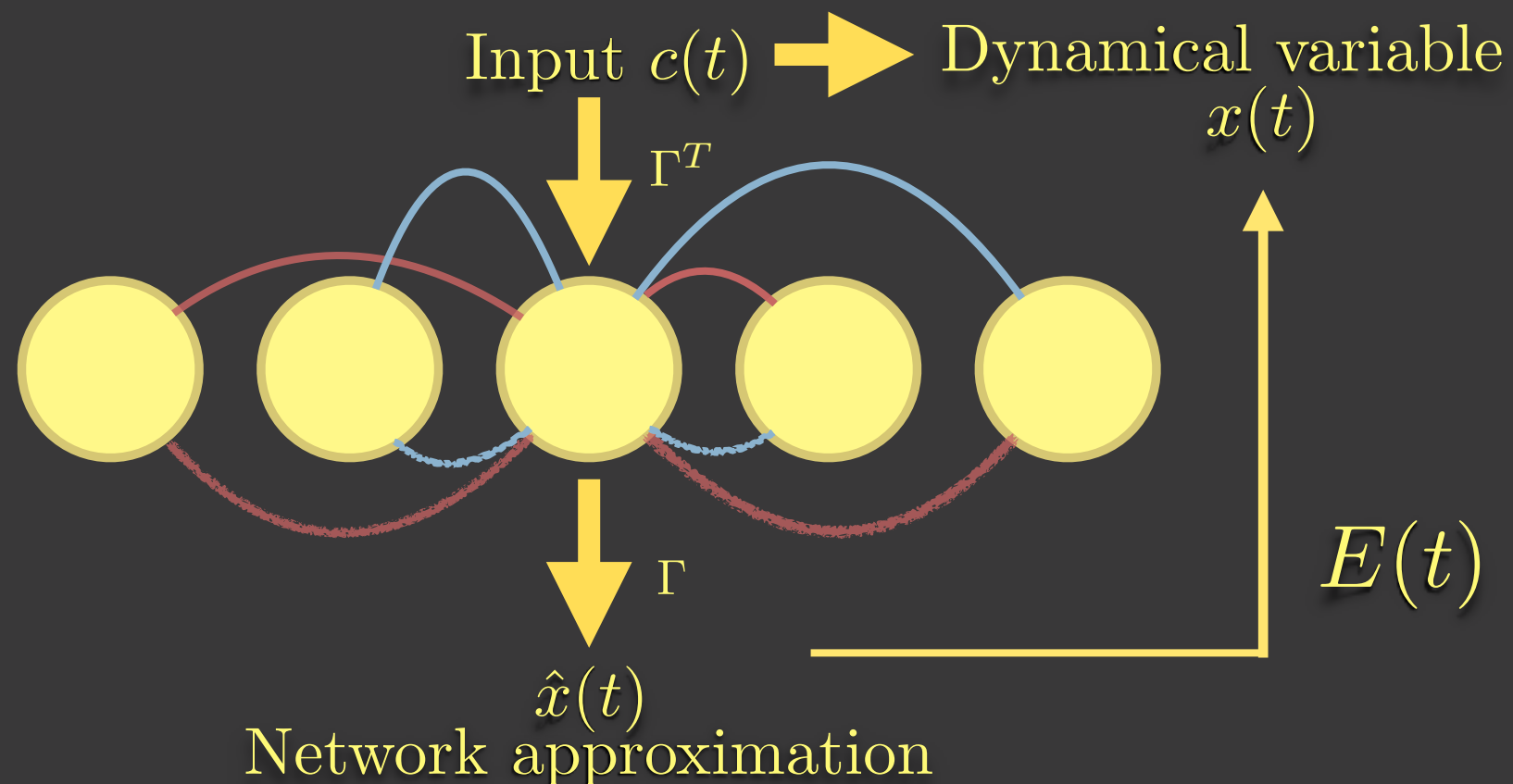
APPLICATION : OPTIC TECTUM TUNING CURVES / RFs



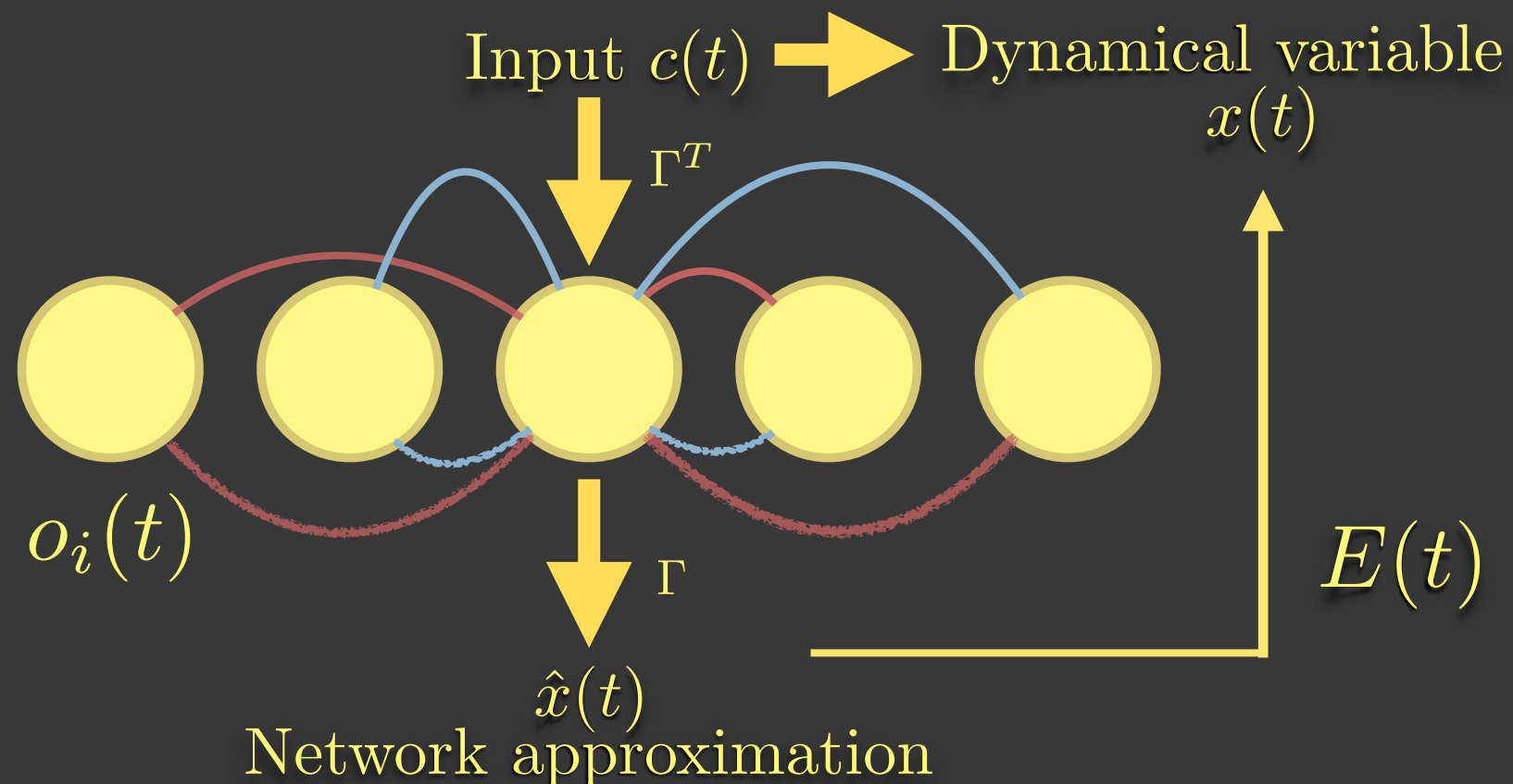
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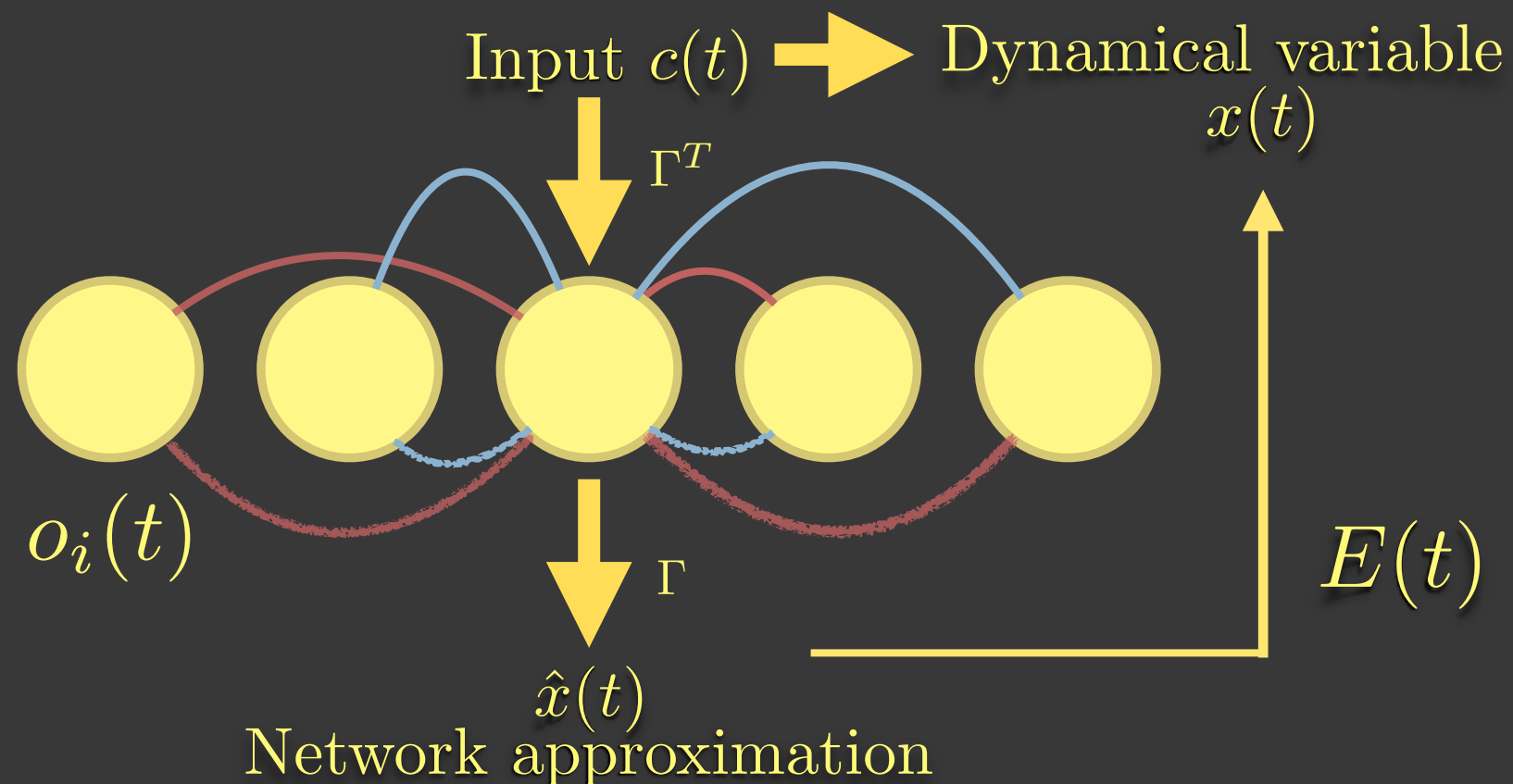
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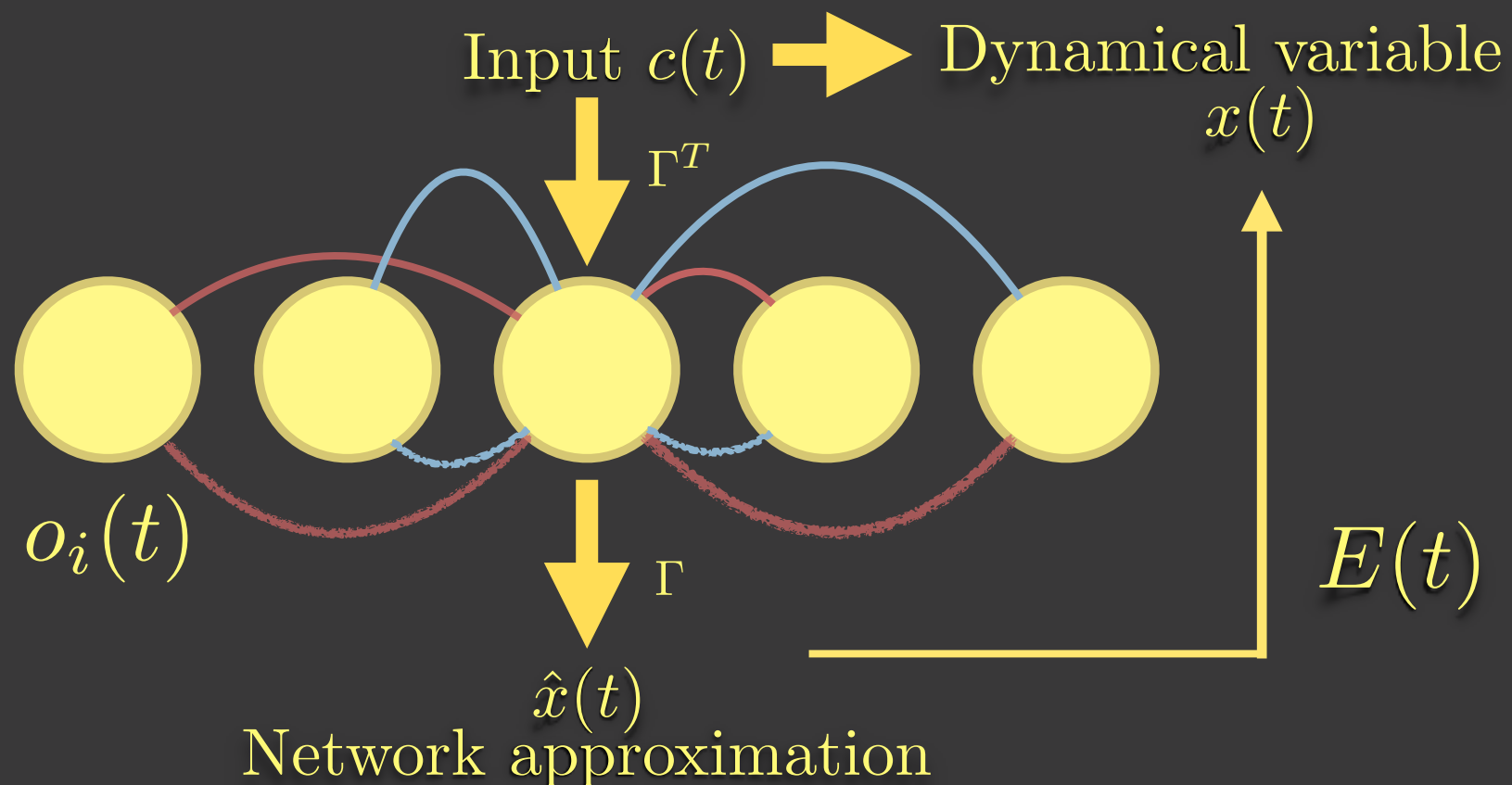
PREDICTIVE CODING IN BALANCED SPIKING NETWORKS



$$o_i(t) = \delta(t)$$

$$\dot{\hat{x}} = -\lambda_d \hat{x} + \Gamma o(t)$$

PREDICTIVE CODING IN BALANCED SPIKING NETWORKS



$$o_i(t) = \delta(t)$$

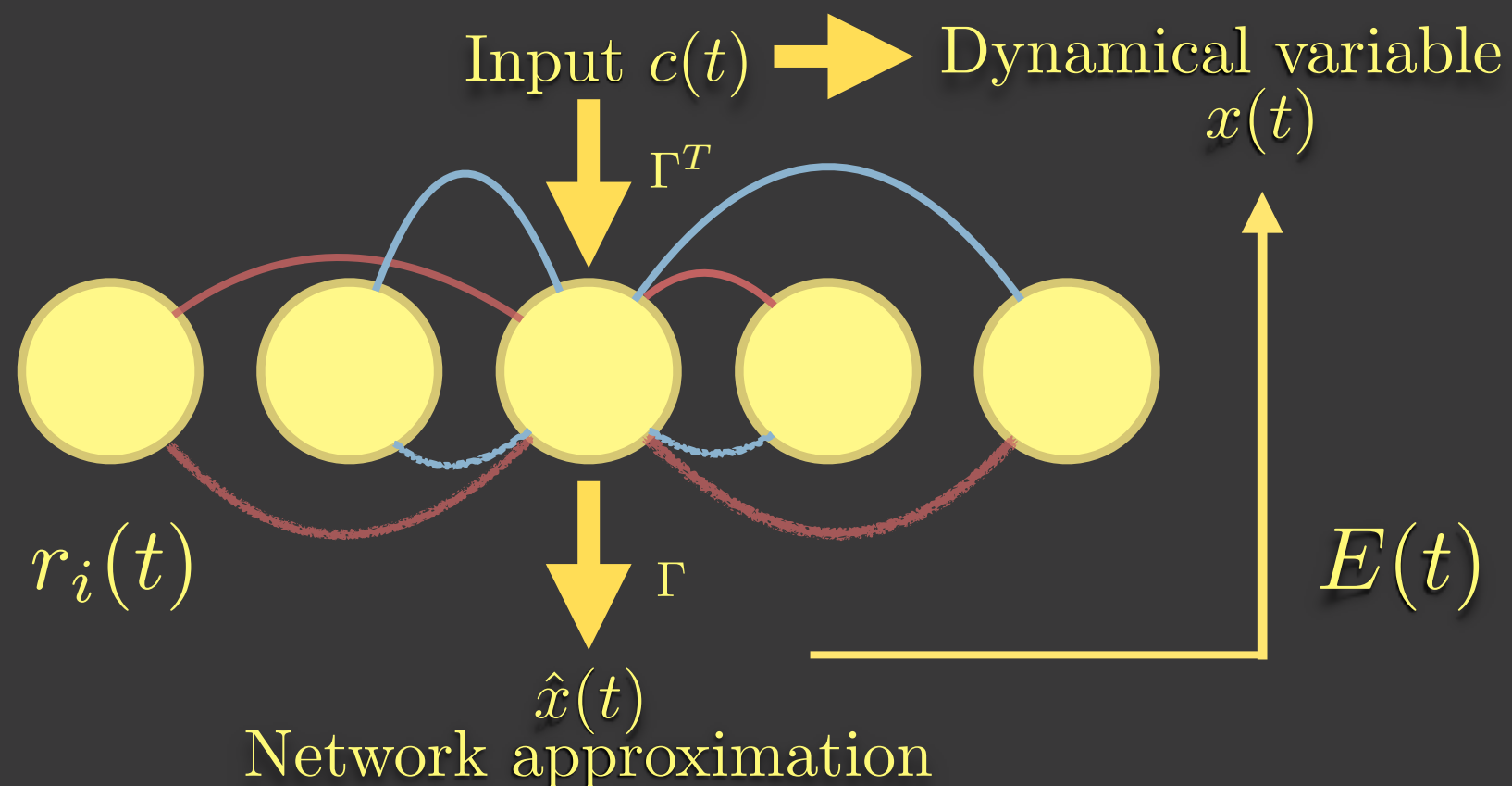
$$\dot{\hat{x}} = -\lambda_d \hat{x} + \Gamma o(t)$$

$$\dot{r} = -\lambda_d r + \lambda_d o(t)$$

time-varying firing rate

PREDICTIVE CODING IN BALANCED SPIKING NETWORKS

$$E(t) = \int_0^t (||x(u) - \hat{x}(u)||_2^2 + \nu ||r(u)||_1 + \mu ||r(u)||_2^2)$$



PREDICTIVE CODING IN BALANCED SPIKING NETWORKS

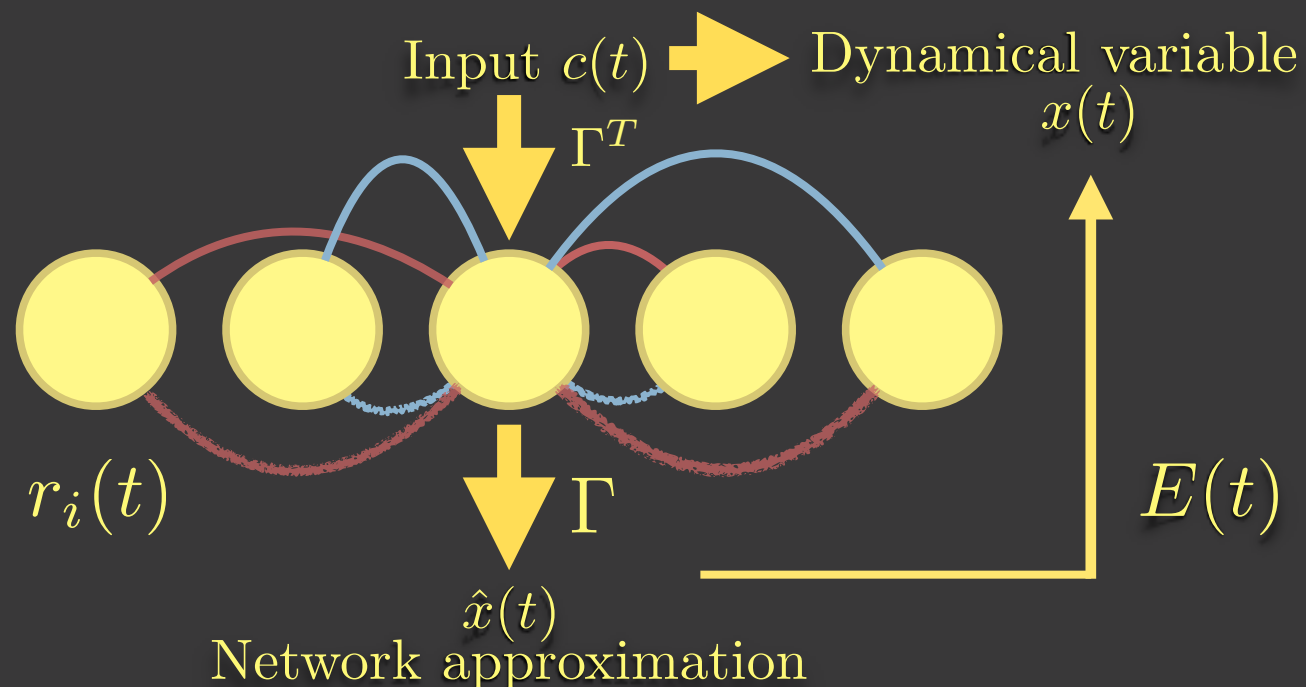
$$E(t) = \int_0^t (||x(u) - \hat{x}(u)||_2^2 + \nu ||r(u)||_1 + \mu ||r(u)||_2^2)$$

$$V_i(t) > T_i$$

$$V_i(t) = \Gamma_i^T (x(t) - \hat{x}(t)) - \mu \lambda_d r_i(t)$$

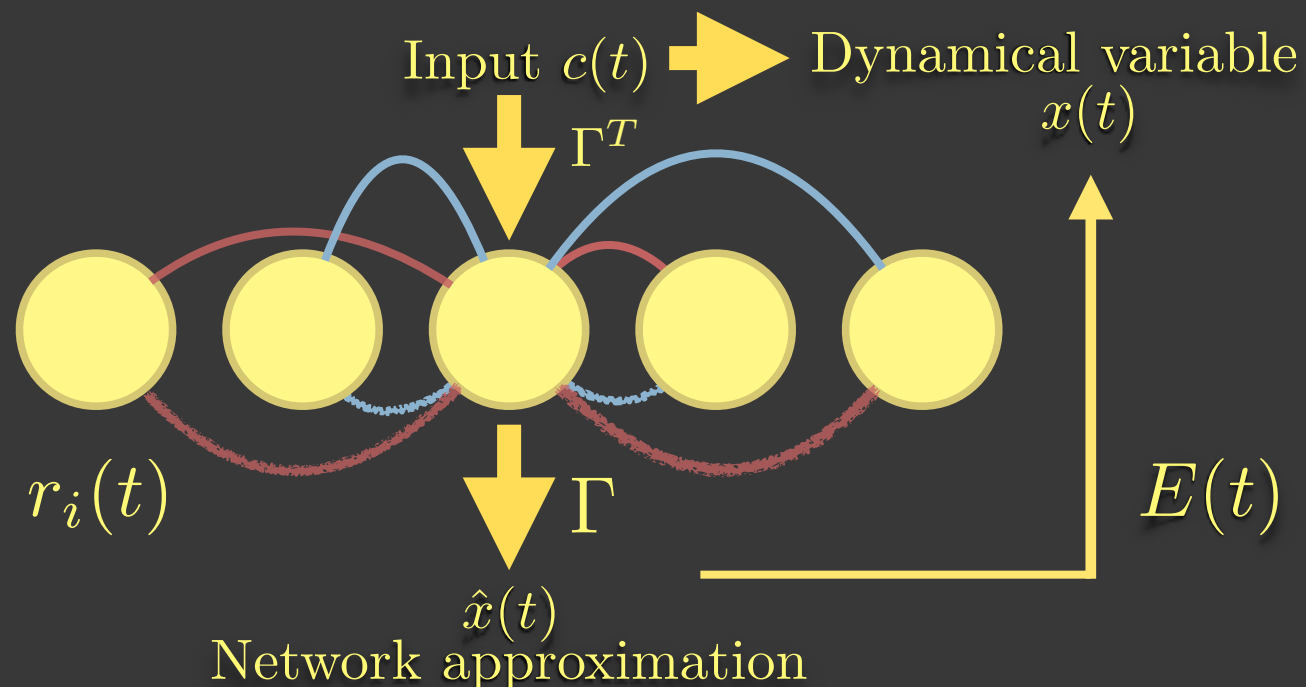
$$T_i = \frac{\nu \lambda_d + \mu \lambda_d^2 + ||\Gamma_i||^2}{2}$$

λ_d read-out's decay rate.



PREDICTIVE CODING IN BALANCED SPIKING NETWORKS

$$E(t) = \int_0^t (||x(u) - \hat{x}(u)||_2^2 + \nu ||r(u)||_1 + \mu ||r(u)||_2^2)$$



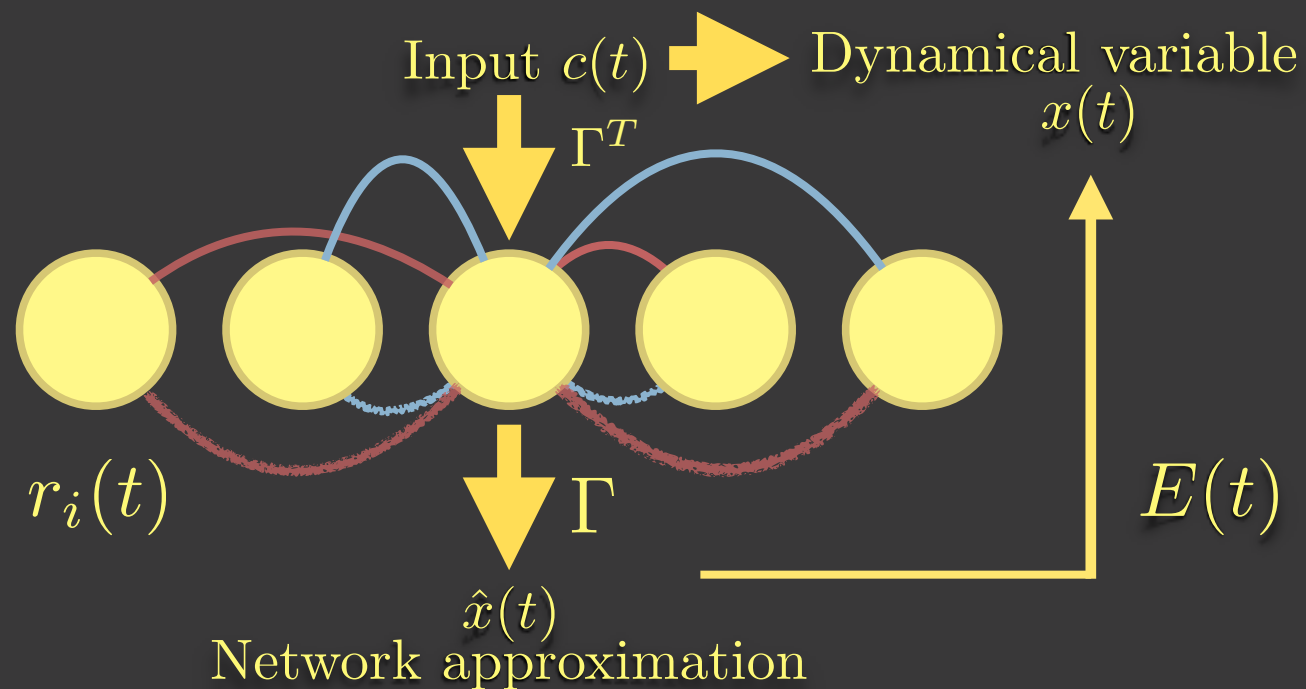
$$V_i(t) > T_i$$

$$o_i(t) = \delta(t)$$

- $\dot{\hat{x}} = -\lambda_d \hat{x} + \Gamma o(t)$
- $\dot{r} = -\lambda_d r + \lambda d o(t)$
- $E(t)$ updated

PREDICTIVE CODING IN BALANCED SPIKING NETWORKS

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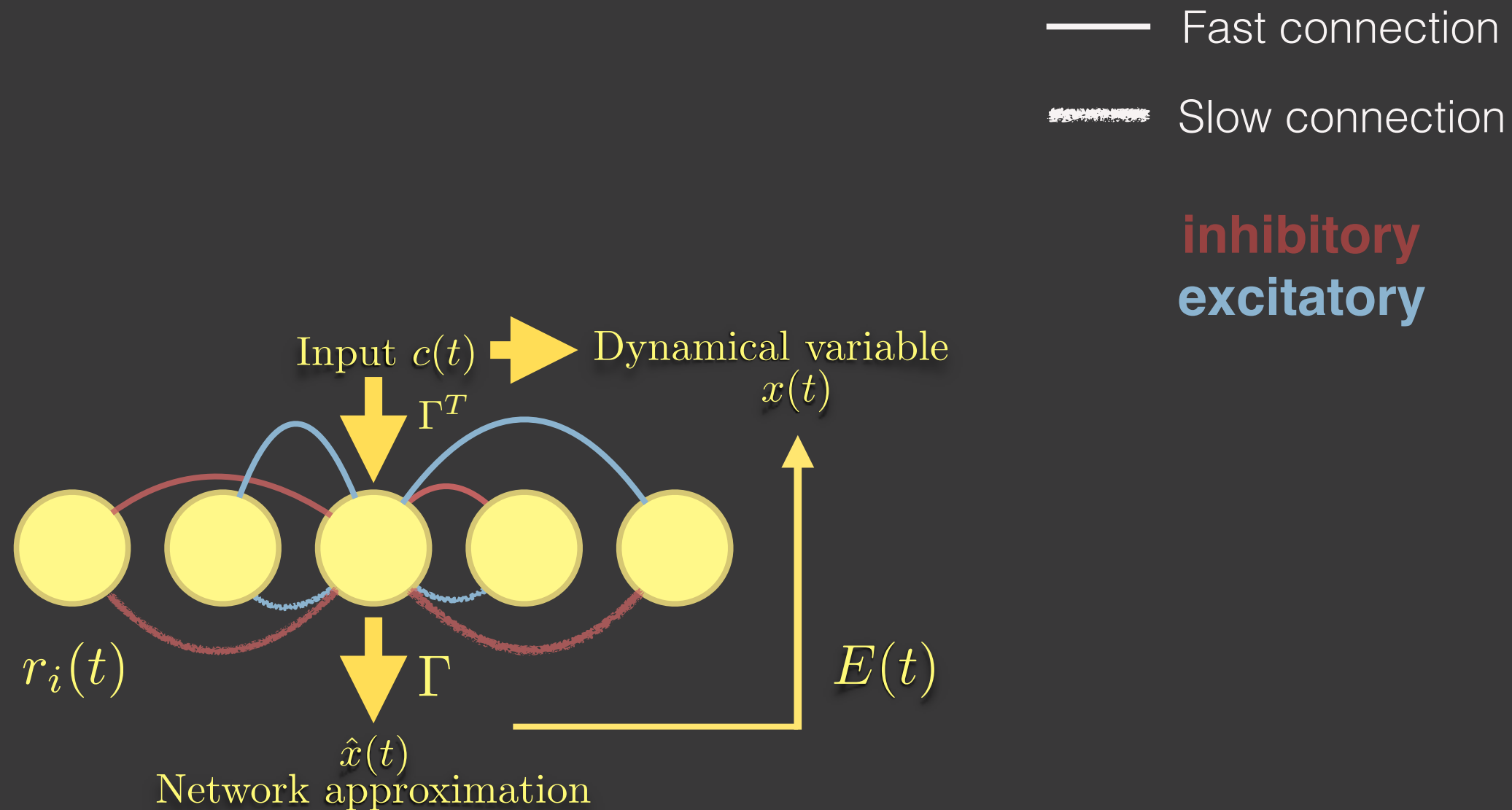
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x dynamics:

$$-\dot{x} = -\lambda_s x + c(t)$$

PREDICTIVE CODING IN BALANCED SPIKING NETWORKS

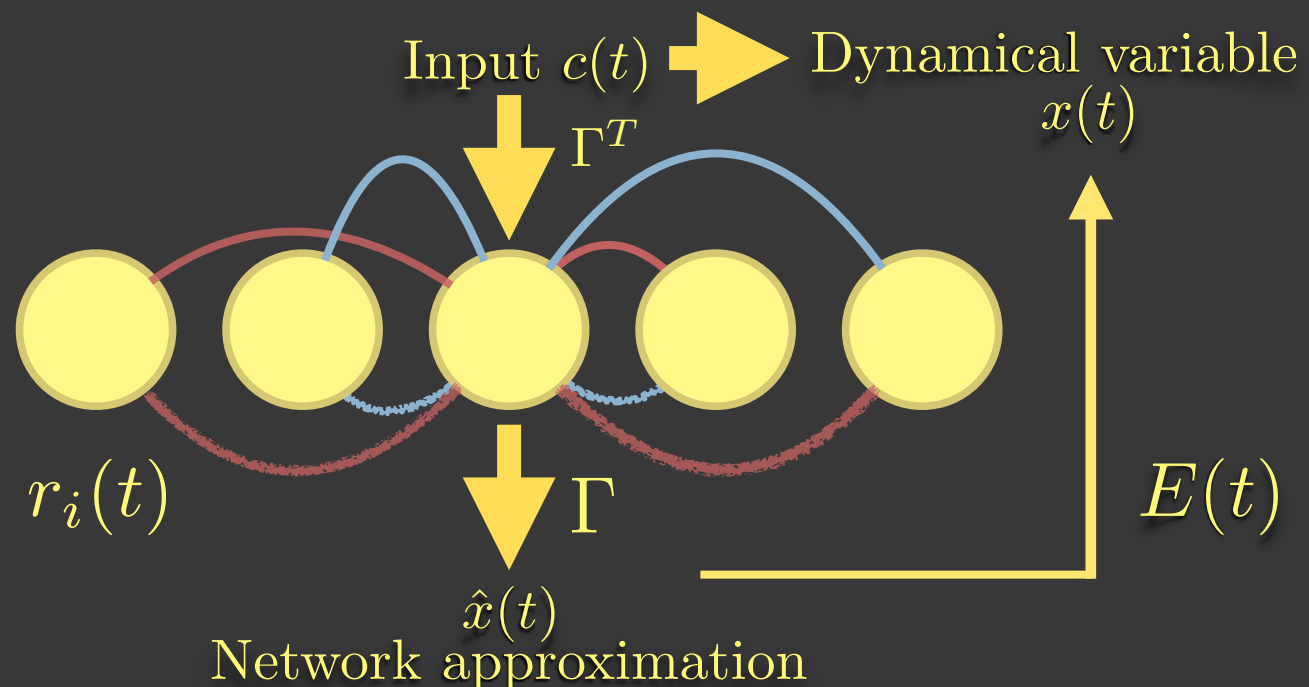


PREDICTIVE CODING IN BALANCED SPIKING NETWORKS

Γ
 λ_d read-out's decay rate.

— Fast connection
 - - - Slow connection

inhibitory
excitatory



PREDICTIVE CODING IN BALANCED SPIKING NETWORKS

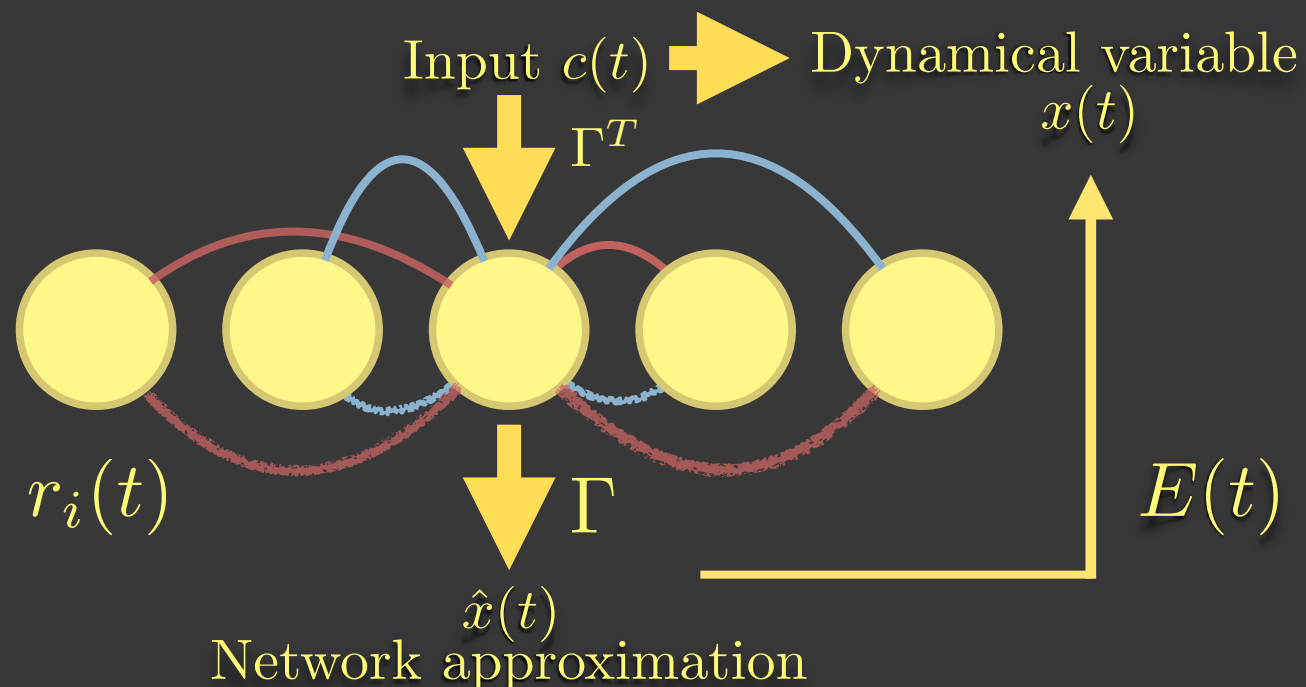
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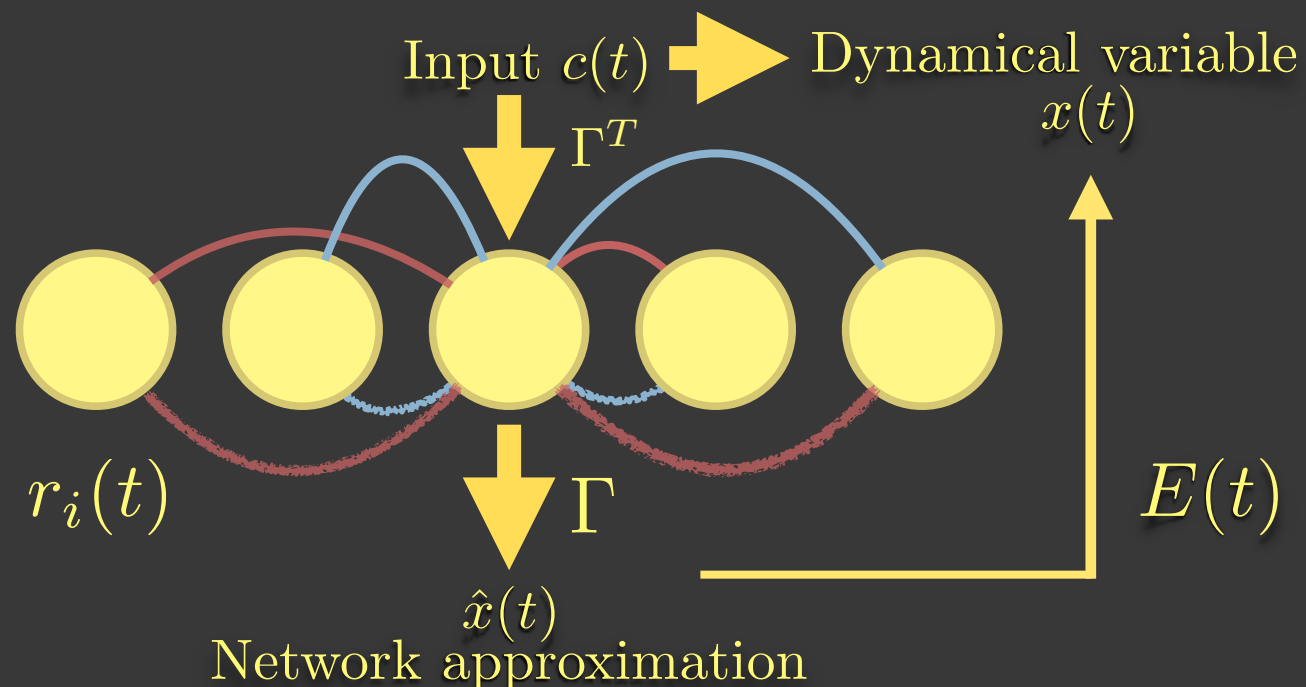
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— Fast connection $\Omega^f = \Gamma^T \Gamma + \mu \lambda_d^2 I$

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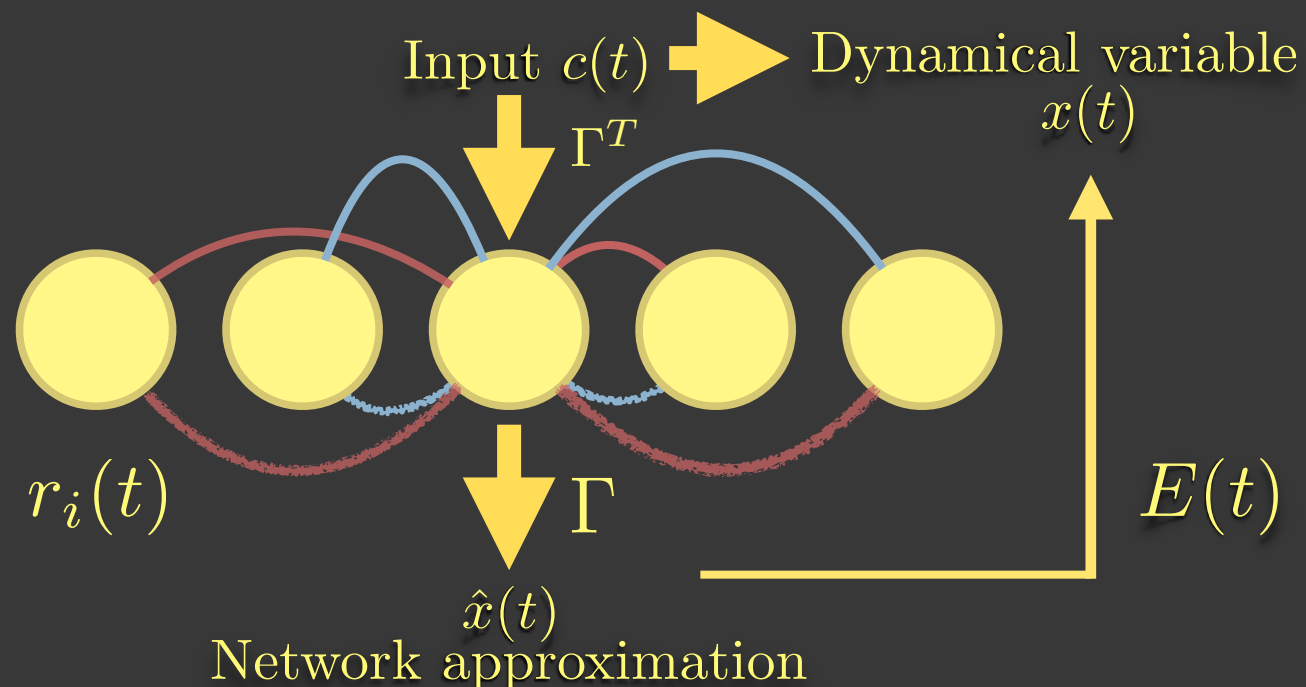
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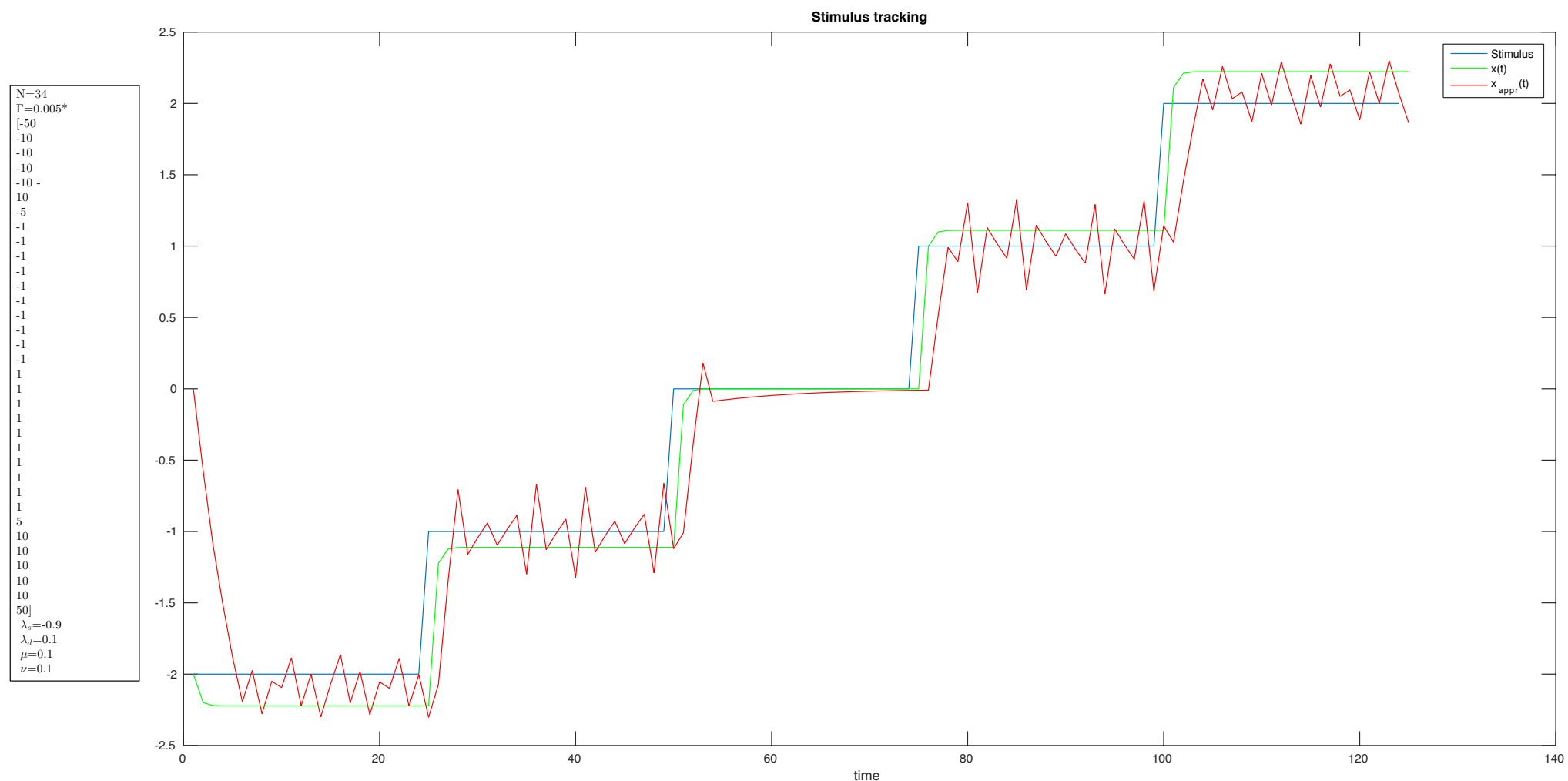


effective connectivity:

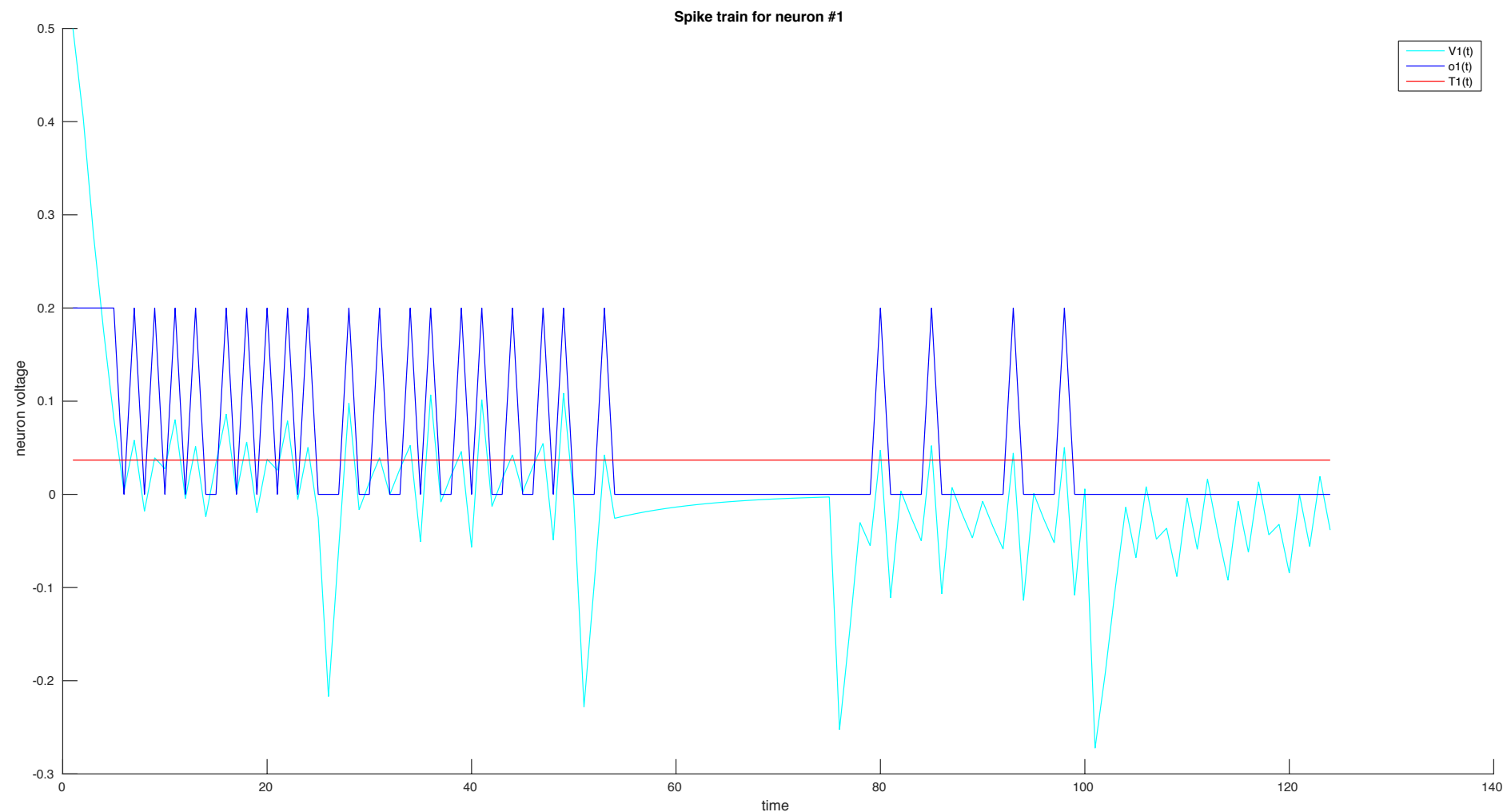
$$\dot{V}_i = -\lambda_v V_i + \sum_{k=1}^{N_{neurons}} W_{ik} * o_k(t) + \Gamma_i^T c(t)$$

$$W_{ik}(t) = \Omega_{ik}^s h_d(t) - \Omega_{ik}^f \delta(t)$$

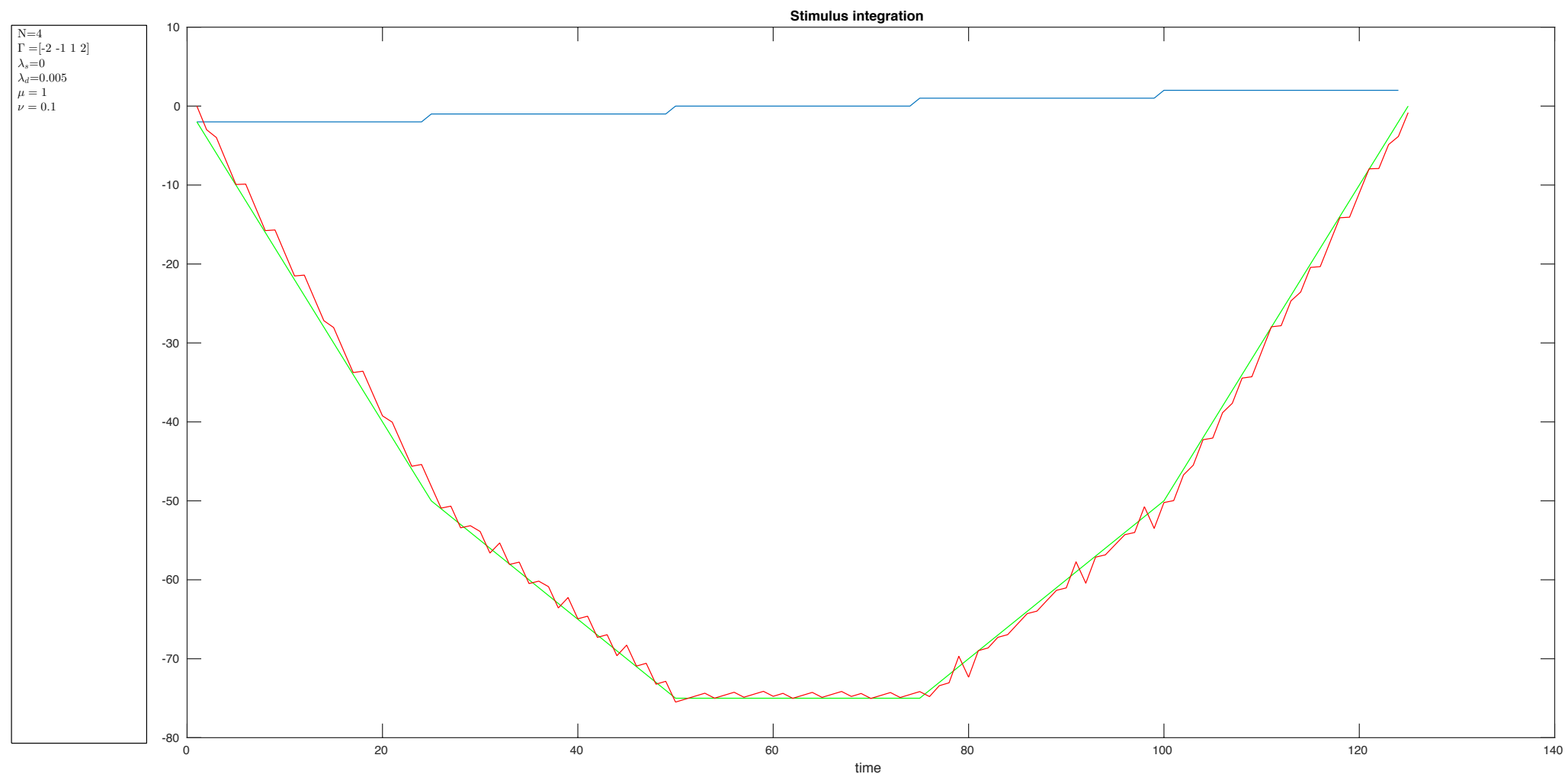
PREDICTIVE CODING : STIMULUS TRACKING EXAMPLE



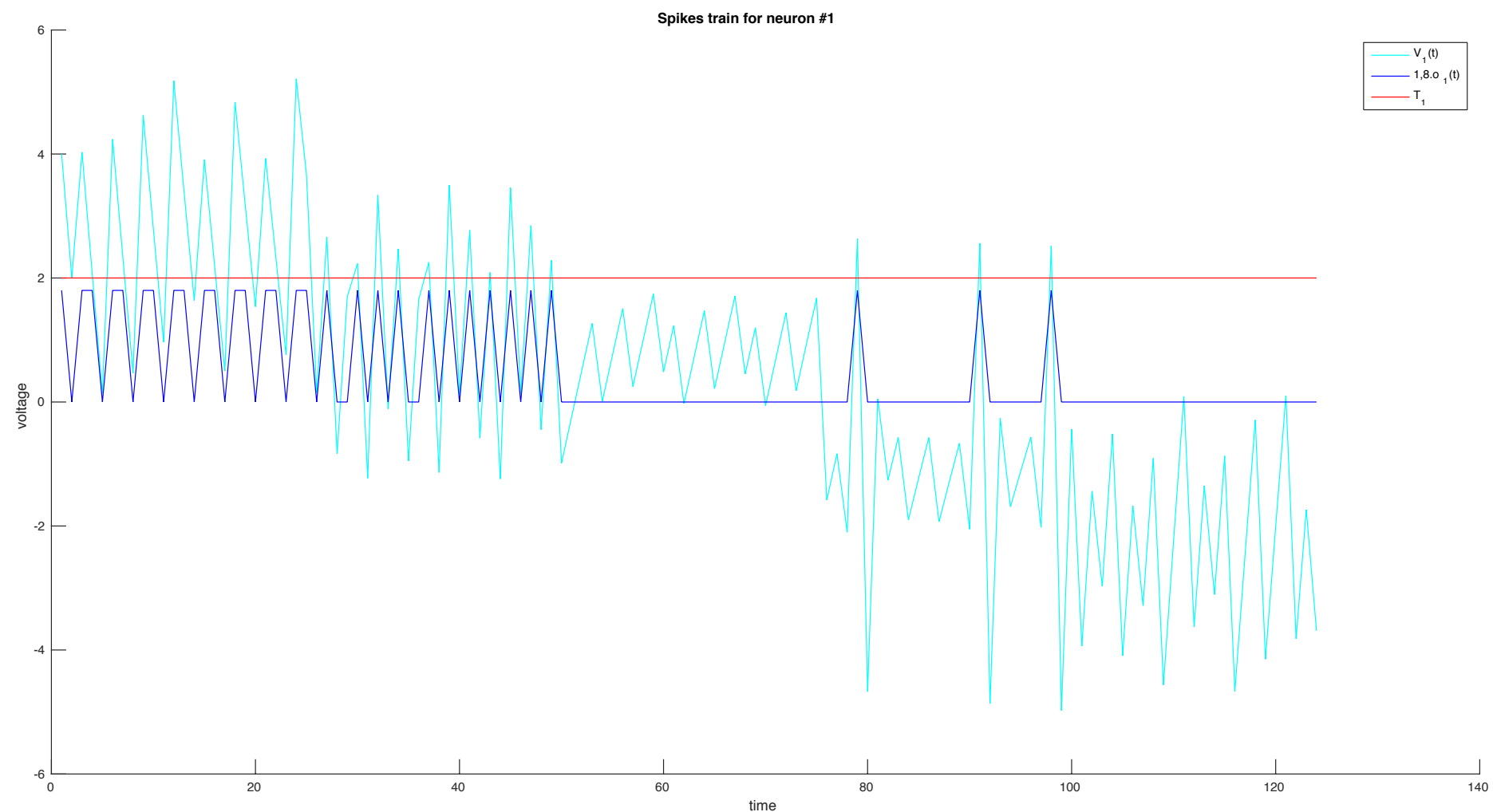
PREDICTIVE CODING : STIMULUS TRACKING EXAMPLE



PREDICTIVE CODING : INTEGRATOR EXAMPLE



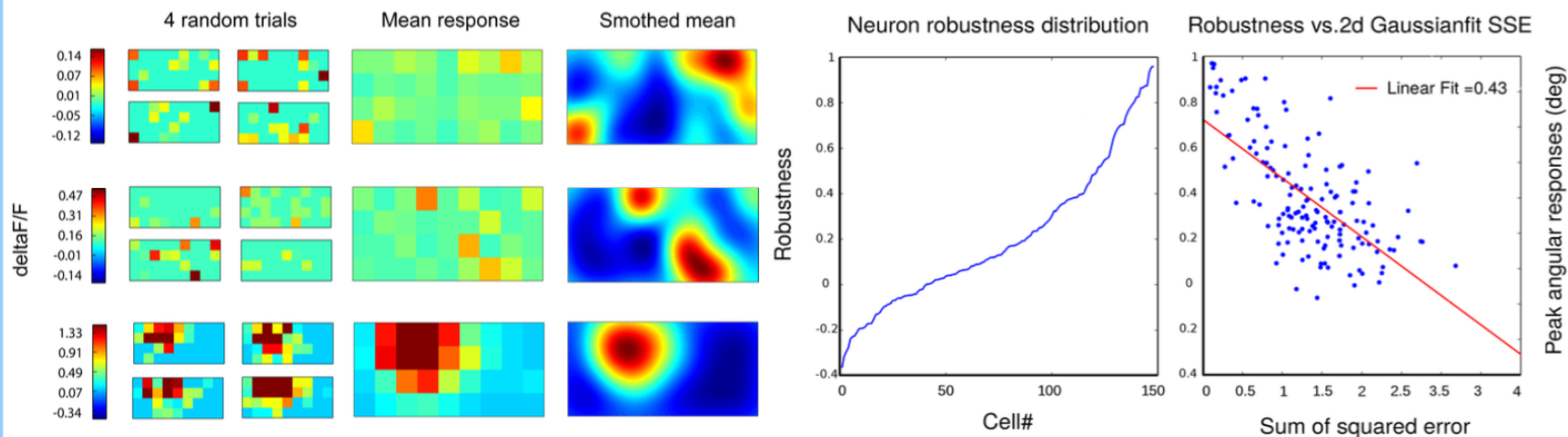
PREDICTIVE CODING : INTEGRATOR EXAMPLE



OPTIC TECTUM TUNING CURVES

Visual neurons in Optic Tectum show well shaped visual field specificity or direction selectivity. They are organised in a retinotopic manner.

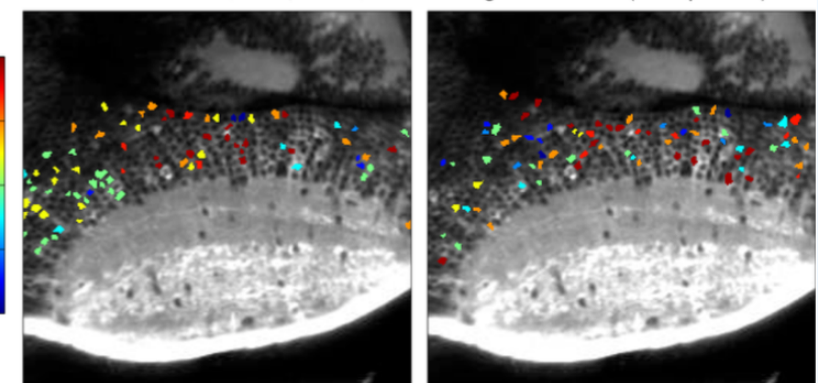
Robust tectal neurons are tuned towards specific regions of the visual field.



$$\text{Robustness}(i) = \text{corr}(\mu(T(i)_{i-k}), \mu(T(i)_{i-m}))$$

Low Sse Cells (first quartile)

High Sse Cells (last quartile)



Neurons display broad range of robustness (defined as correlation in between stimulation trials) for squares responses. Robust neurons displayed low sum of squared errors (SSE) when fit to a 2D gaussian function indicating that these neurons were tuned toward specific regions of the visual field. However, neurons that are poorly tuned might be sensitive to other visual stimuli or sensory modalities.

Robust neurons were scattered across the entire tectum and their peak angular response was organised in a retinotopic manner. In contrary, unrobust neurons were not organised in a retinotopic manner.

OPTIC TECTUM TUNING CURVES

Visual neurons in Optic Tectum show well shaped visual field specificity or direction selectivity.
They are organised in a retinotopic manner.

But some of them show noisy response for squares or moving bar stimulus, showing no clear receptive field attribution or direction selectivity.
What about them ?

Idea : implement a network with the same number of neurons as recorded entities.
Look for stable representation through spike trains for the ones known as stable, investigate coding role of the remaining neurons.

OPTIC TECTUM TUNING CURVES

Pb : how to tune network's characteristics ?
More precisely : what about kernels' choice ?

Suggestion : learning Γ from the data using a classical linear regression.

Find W such that:

$$r = W.x$$

where:

x is the input's descriptive variable (position, angle)

$$r \in \mathbb{R}^{N_{neurons} \times T}$$

State $\Gamma_i = f(W_i \in \mathbb{R}^{N_{features}(x)})$

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Underlying idea : when error projection is big enough on one neurone kernel, the neurone spikes.

So Γ can be seen as a set of vectors of preferred directions for network's correction game.

Each neurone has its preferred direction.

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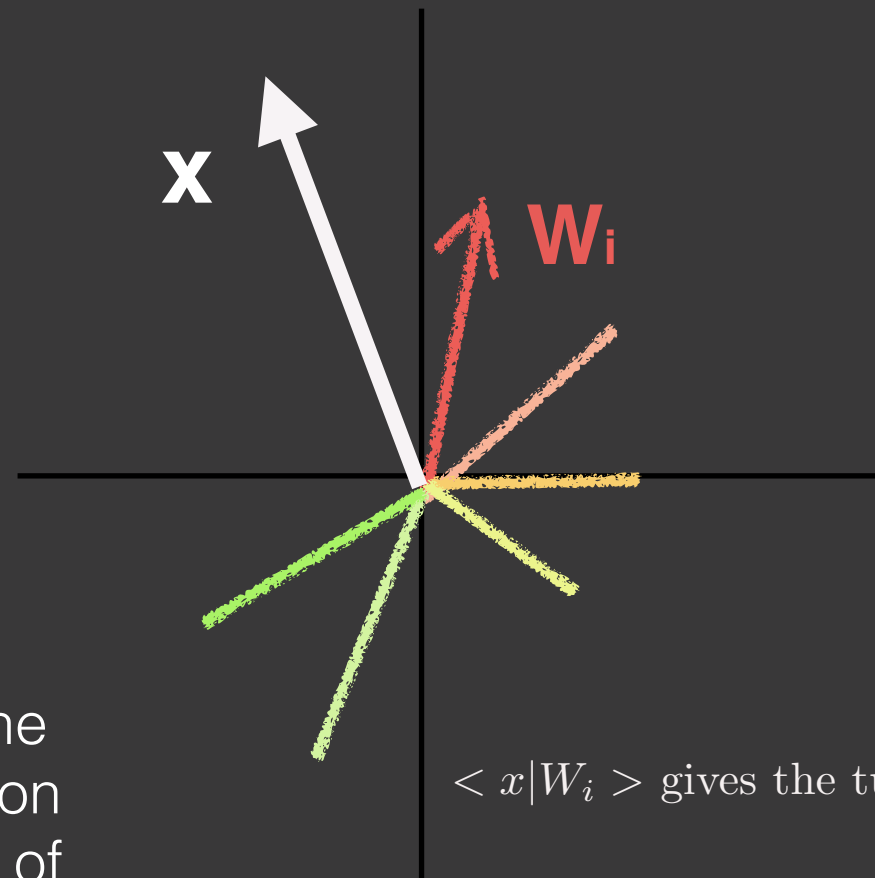
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#####

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$\langle x | W_i \rangle$ gives the tuning curve for **neurone i**

Tuning curve for neurone i should be given by the scalar product of its theoretical preferred direction (according to the network) with the whole range of stimulus.

CONCLUSION

To do

See how a hand-adjusted network decode stimulus for moving bars.
Look at the tuning curves for each Neural Network neuron and see if direction selectivity emerge.

Explicit a mathematical link between W and Γ .

Is there an optimal regression method to infer a 'good' Γ candidate, given the error minimisation objective ?

REFERENCES

Predictive Coding of Dynamical Variables in Balanced Spiking Networks

M.Boerlin, S.Deneve, C.K.Machens

-

The Optic Tectum

D.Northmore

-

Visual Response Properties of Zebrafish Tectal cells

P.Sajovic, C.Levinthal

Neuroscience, 1982

-

Visual Cells of Zebrafish Optic Tectum : Mapping with small dots

P.Sajovic, C.Levinthal

Neuroscience, 1982

-

Functional Elimination of Excitatory Feedforward Inputs Underlies Developmental Refinement of Visual Receptive Fields in Zebrafish

Zhang et al.

The Journal of Neuroscience, 2011

-

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IBENS

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